

Aiden Cullo

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EXPERIENCE

Tech To The Rescue | Lead Software Engineer Sep 2025—

- Building a pipeline to ingest distributed **RTSP** streams, perform real-time processing, and cache video for low latency.
- Orchestrating application deployment onto **AWS EC2** machines using **Terraform** modules and **Docker** containers.
- Building dashboard **UI** that polls water station sensor readings, alerting administrators of anomalies or inconsistencies.
- Constructing a backend server to process registration of new water supply sites, facilitating expansion to more locales.

JPMorgan Chase | Software Engineer II Sep 2024—Mar 2025

- Built a dashboard capable of visualizing 1k+ **LLMs**, allowing data scientists to monitor **ML** workflows.
- Augmented **FastAPI** logs with user IDs across all endpoints, ensuring traceability of 1M+ daily events.
- Validated client identities accessing **LLM** artifact registry, enhancing security and guaranteeing government compliance.
- Watermarked automated reports for non-compliant **LLM** projects to prevent unauthorized sharing to external parties.

HeroGrow | Software Engineer I Apr 2023—Sep 2024

- Developed a **Ruby on Rails** REST API to manage venue seating layouts, allowing real-time ticket reservations.
- Integrated **JavaScript** Turbo Frames into the browser UI, allowing partial page updates, reducing load times by 50%.
- Implemented OAuth2-based authentication, streamlining sign-in processes and increasing user registrations.
- Integrated Square payments API into the customer checkout form, allowing customers to pay quickly and efficiently.

Tradigro | Software Engineer I Dec 2022—Apr 2023

- Indexed **Vue.js** journals, trades, and achievements views, allowing more granular and searchable public content.
- Decoupled raw trade and statistical analysis modules, reducing duplicate feed publications upon user profile updates.
- Standardized text encoding format in **Vuex** frontend data store, eliminating incorrect comment representations.
- Refactored the data upload protocol, serializing I/O on trade files, preventing information loss from race conditions.

SuperCerebros | Junior Software Engineer May 2021—Dec 2022

- Built a progressive web application in **React**, deployed to 10k+ domestic and international users.
- Integrated mobile and desktop applications via Ionic SDK and **TypeScript**, cutting total server requests in half.
- Architected DBMS with **Google Firebase**, storing 1TB of user data while maintaining ms latency.

SUNY Binghamton | Research Assistant Aug 2018—Nov 2020

- Tuned evolutionary search algorithm for nanocrystal structures in **C** to obtain stabler ground states 90% of the time.
- Achieved 50% speedup of the open source **MAISE** library via parallelization with **OpenMP** toolkit.
- Automated neural network training with **Bash** scripts, reducing the iteration cycle from 100 to 20 hours.

Caltech | Physics Fellow Jun 2019—Aug 2019

- Performed scanning tunneling microscopy of topological defects in $(\text{BiSb})_2\text{Te}_3$ -bi-layer topological insulator substrates.

Sapienza University of Rome | Software Engineer Intern Jun 2018—Aug 2018

- Built a multi-layer convolutional neural network in **Python**, classifying particle collisions with 99.5% accuracy.
- Generated 20k training samples based on weakly interacting massive particle (WIMP) theoretical models.

PROJECTS

pipenv | Contributor Dec 2023—Feb 2024

- Migrated Click library to Rich and argparse tooling, reducing the package size by 1MB and cutting vendor dependencies.

SKILLS

Programming Languages:	Python, JavaScript, TypeScript, C++, C, Java, Ruby, SQL, Go, Haskell, PHP, Rust
Web:	React, Vue.js, Angular, Node.js, Ruby on Rails, FastAPI, Flask, Django, Next.js, Express, Dart
Databases:	PostgreSQL, MySQL, SQLite, Firebase, Redis, Cassandra, Oracle, DynamoDB, MongoDB
Tools:	Git, Docker, Kubernetes, Jenkins, Apache, Nginx, AWS, Azure, GCP, Heroku, Netlify, Vercel
Spoken Languages:	English (<i>native</i>), Spanish (<i>fluent</i>), French (<i>proficient</i>), Portuguese (<i>basic</i>), Mandarin (<i>basic</i>)

EDUCATION

SUNY Binghamton Class of 2020

- BS Computer Science, *Watson School of Engineering*
- BS Mathematics, *Harpur College of Arts and Sciences*
- BS Applied Physics, *Harpur College of Arts and Sciences*